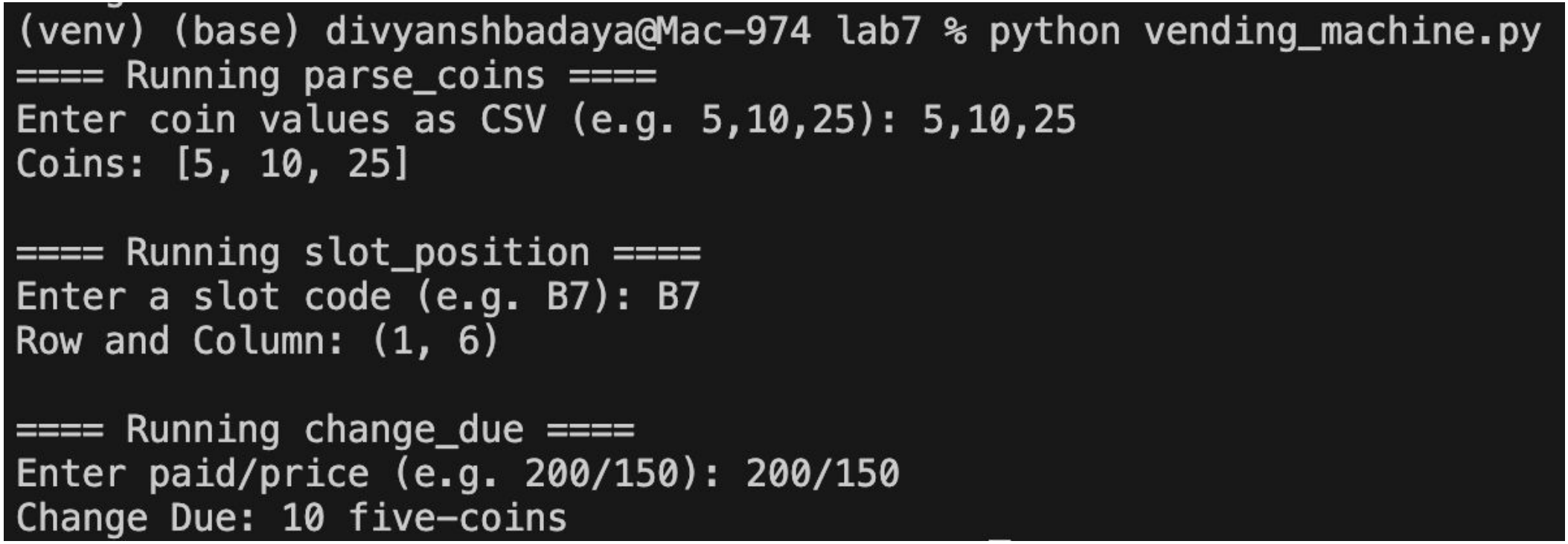
Lab 7 – Fuzz Testing Evidence

Vending Machine Application | Divyansh Badaya | PES1UG24AM094

This document contains the submitted terminal screenshots showing application behaviour, Hypothesis-discovered failures in the original implementation, and the passing tests and valid-input output after fixes.

Screenshot 1



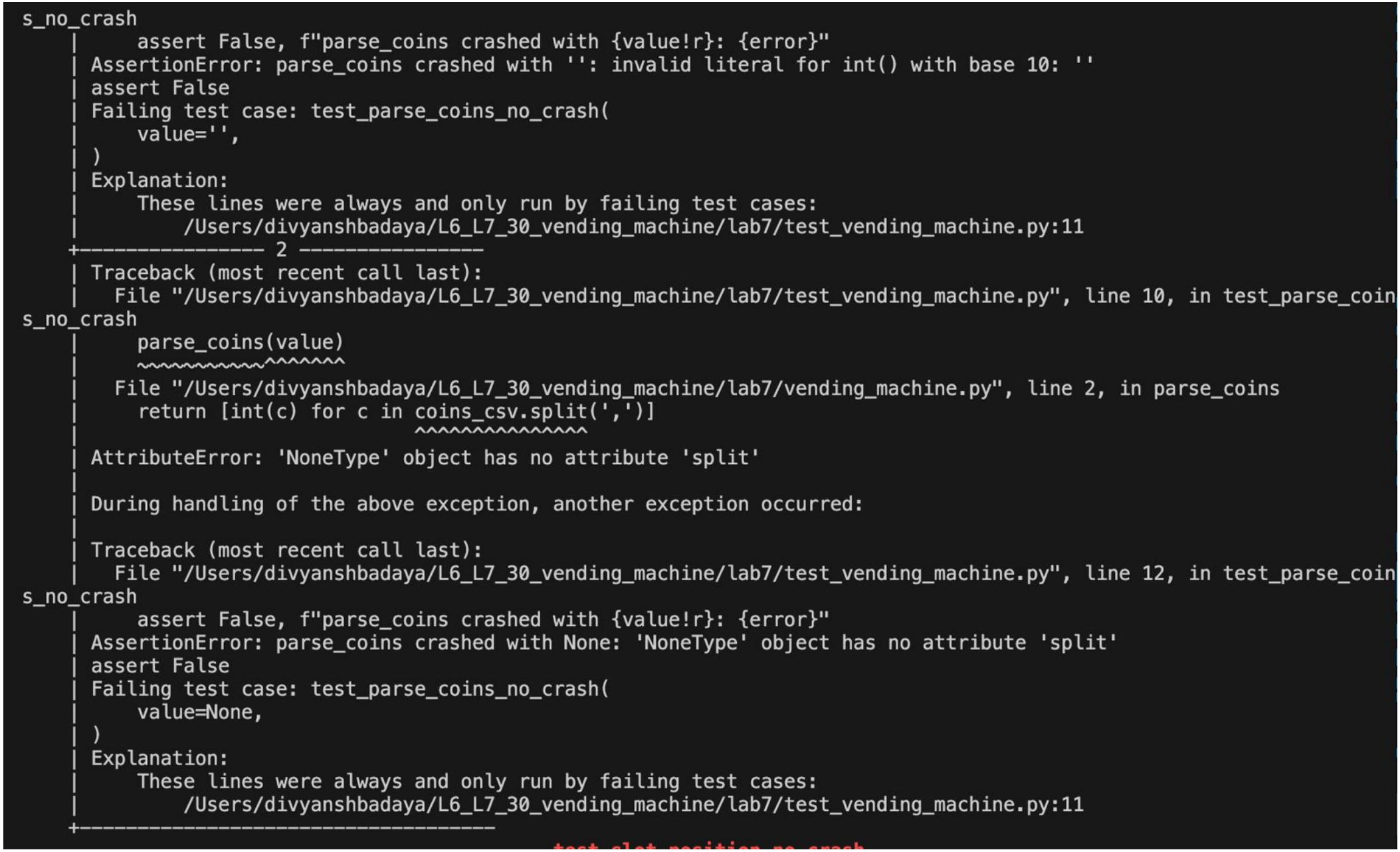
Screenshot 2

```
(venv) (base) divyanshbadaya@Mac-974 lab7 % python -m pytest test_vending_machine.py -v
============================= test session starts ==============================
platform darwin -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0 -- /Users/divyanshbadaya/L6_L7_30_vending_machine/lab6/venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/divyanshbadaya/L6_L7_30_vending_machine/lab7
plugins: hypothesis-6.170.15
collected 3 items

test_vending_machine.py::test_parse_coins_no_crash FAILED                [ 33%]
test_vending_machine.py::test_slot_position_no_crash FAILED              [ 66%]
test_vending_machine.py::test_change_due_no_crash FAILED                 [100%]

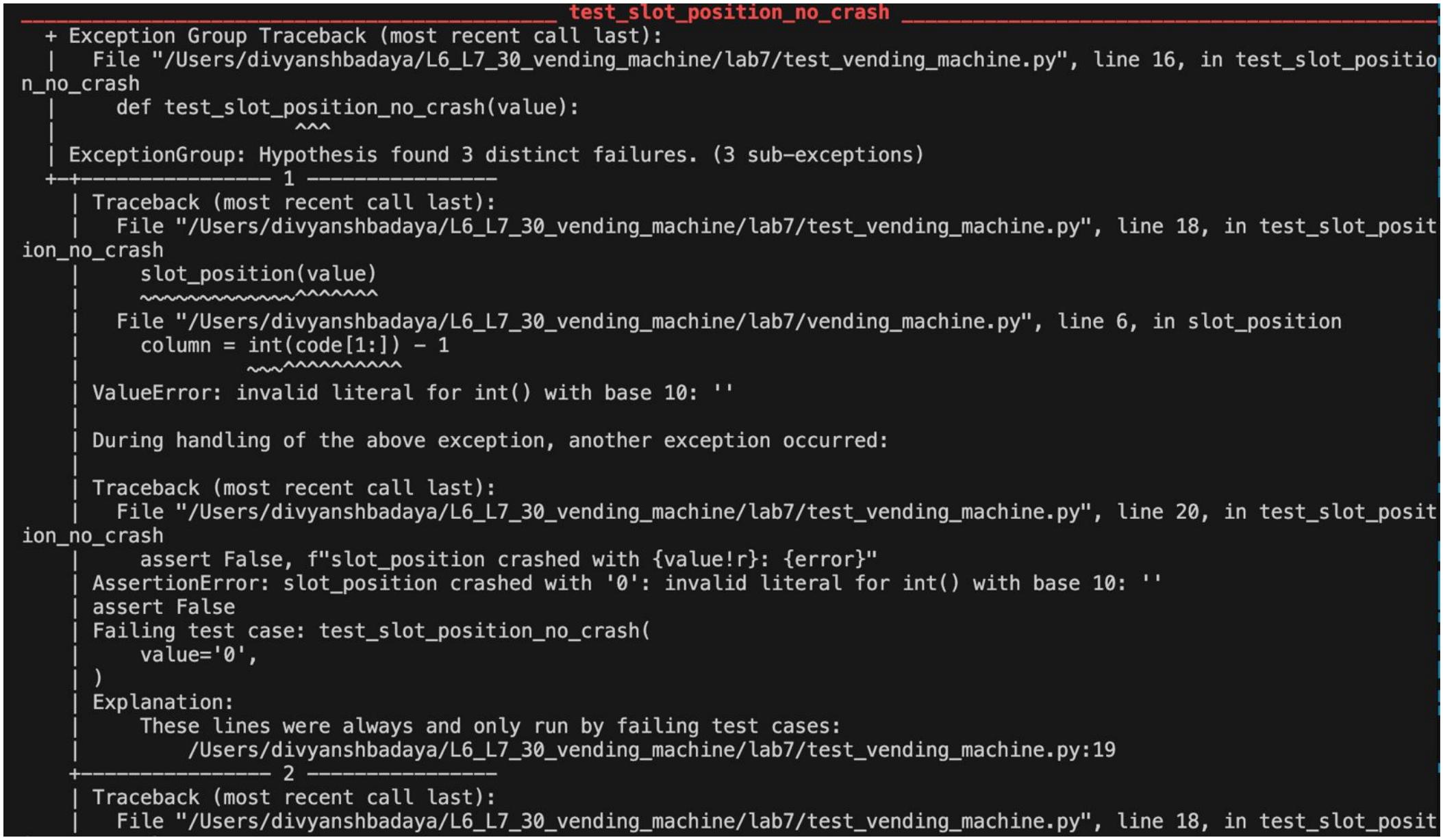
=================================== FAILURES ===================================
_____________________________ test_parse_coins_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 8, in test_parse_coins_no_crash
  |     def test_parse_coins_no_crash(value):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 10, in test_parse_coins_no_crash
    |     parse_coins(value)
    |     ~~~~~~~~~~~^^^^^^^
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/vending_machine.py", line 2, in parse_coins
    |     return [int(c) for c in coins_csv.split(',')]
    |                ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 12, in test_parse_coins_no_crash
    |     assert False, f"parse_coins crashed with {value!r}: {error}"
    | AssertionError: parse_coins crashed with '': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_coins_no_crash(
```

Screenshot 3



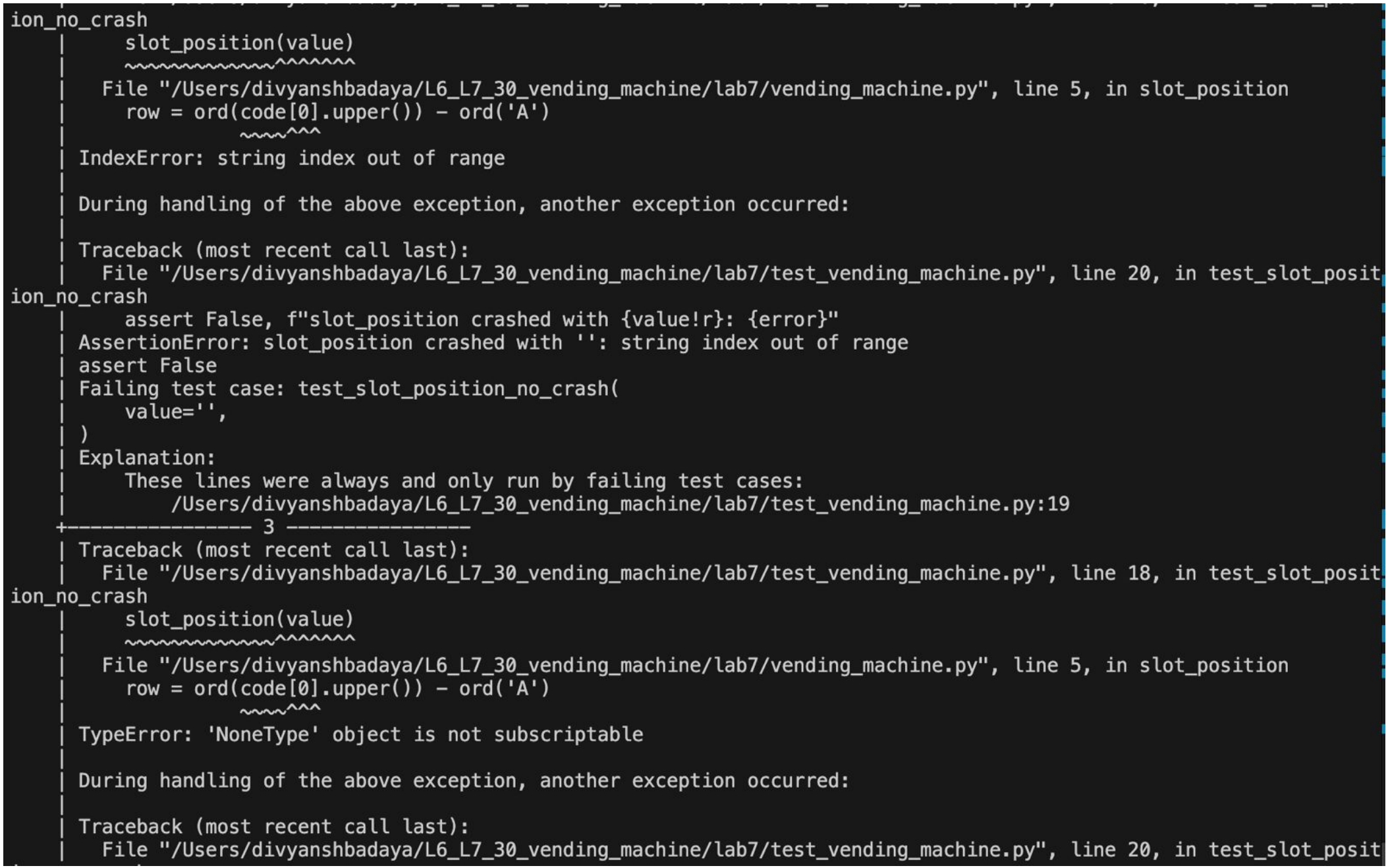
```
s_no_crash
    |     assert False, f"parse_coins crashed with {value!r}: {error}"
    | AssertionError: parse_coins crashed with '': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_coins_no_crash(
    |     value='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py:11
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 10, in test_parse_coin
s_no_crash
    |     parse_coins(value)
    |     ^^^^^^^^^^^^^^^^^^
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/vending_machine.py", line 2, in parse_coins
    |     return [int(c) for c in coins_csv.split(',')]
    |                             ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 12, in test_parse_coin
s_no_crash
    |     assert False, f"parse_coins crashed with {value!r}: {error}"
    | AssertionError: parse_coins crashed with None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_coins_no_crash(
    |     value=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py:11
    +-----------------------------------
                                        test_slot_position_no_crash
```

Screenshot 4



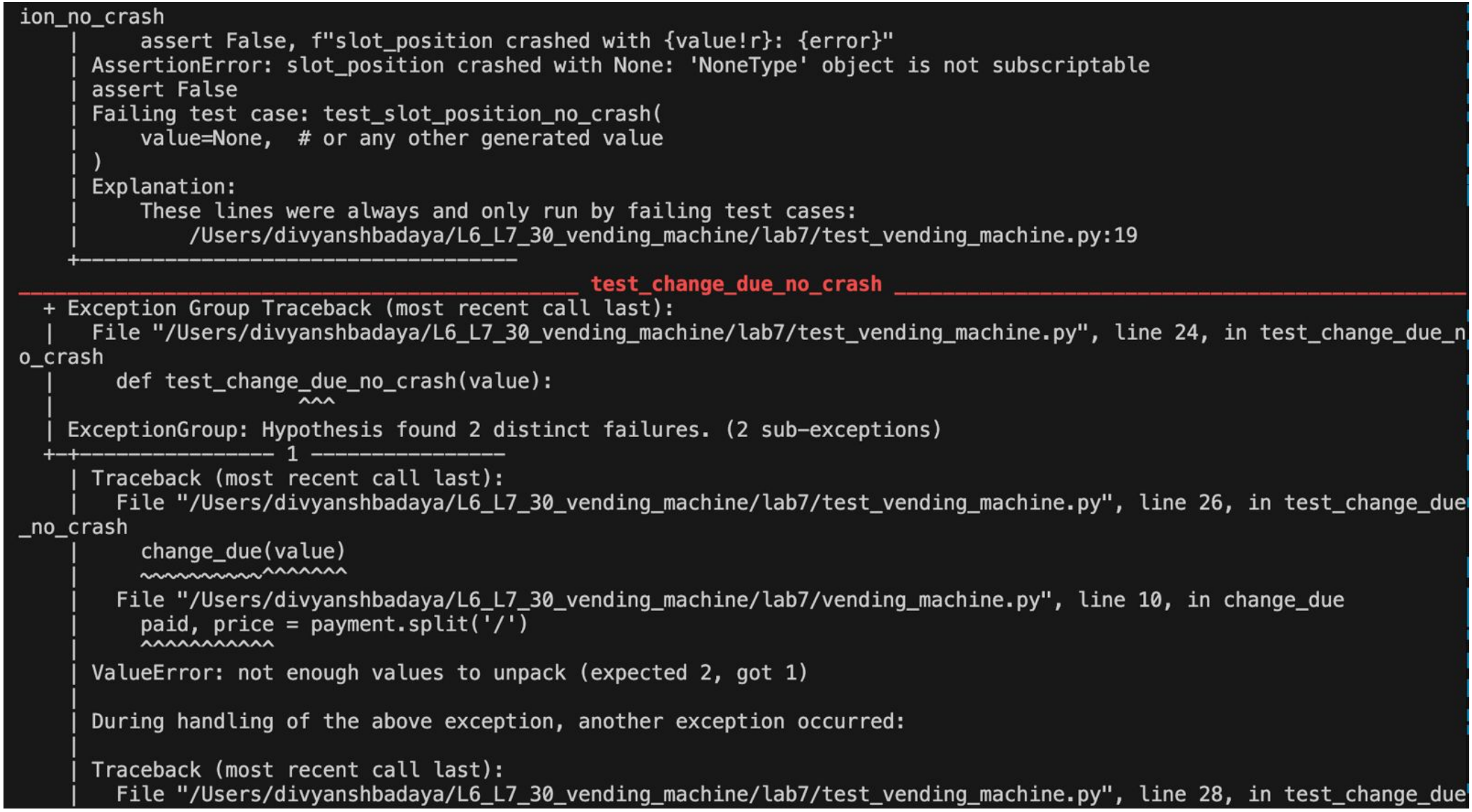
```
_____ test_slot_position_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 16, in test_slot_positio
n_no_crash
  |     def test_slot_position_no_crash(value):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 18, in test_slot_posit
ion_no_crash
    |     slot_position(value)
    |     ~~~~~~~~~~~~~^^^^^^^
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/vending_machine.py", line 6, in slot_position
    |     column = int(code[1:]) - 1
    |              ~~~^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 20, in test_slot_posit
ion_no_crash
    |     assert False, f"slot_position crashed with {value!r}: {error}"
    | AssertionError: slot_position crashed with '0': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_slot_position_no_crash(
    |     value='0',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py:19
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 18, in test_slot_posit
```

Screenshot 5



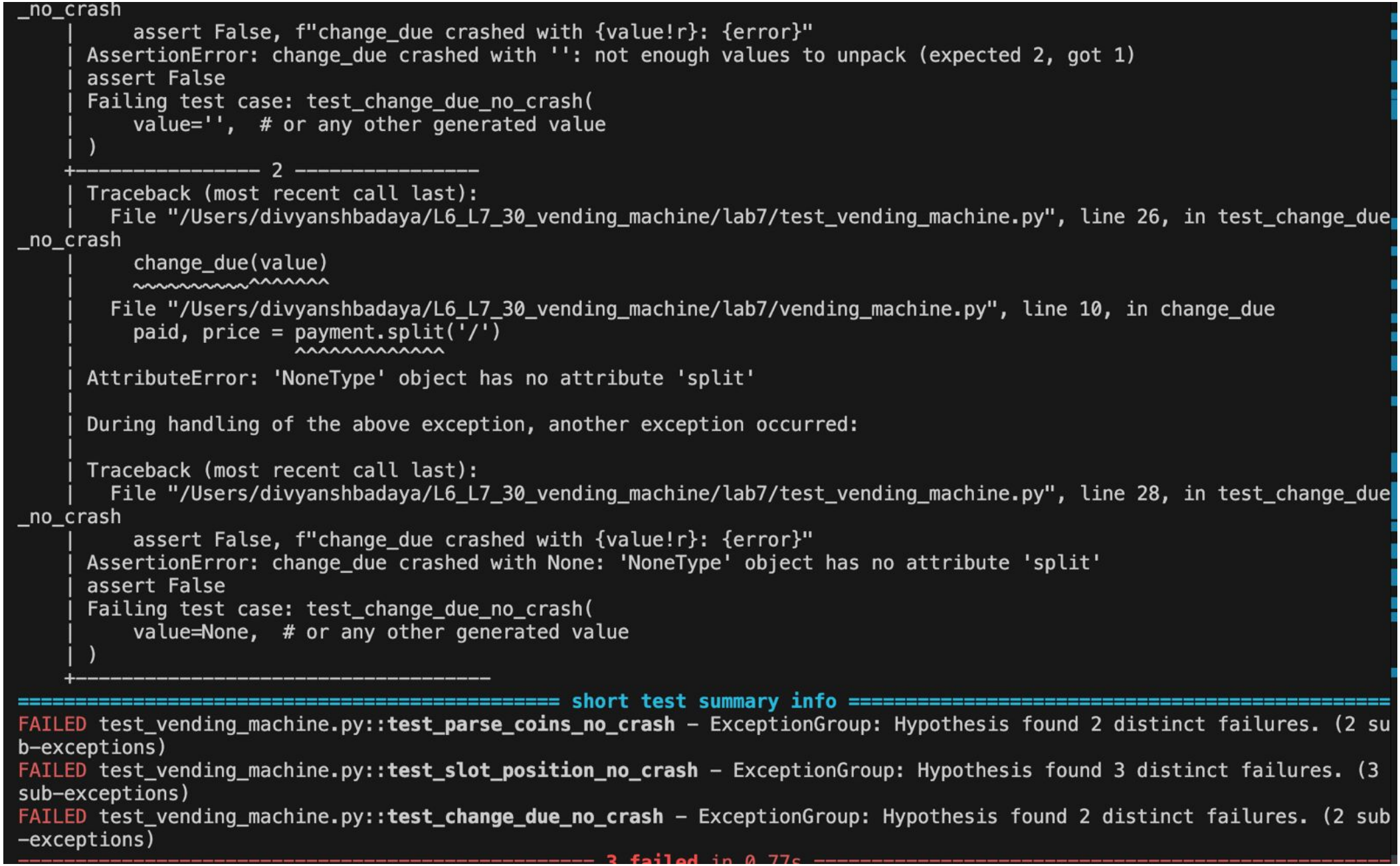
```
ion_no_crash
    |     slot_position(value)
    |     ~~~~~~~~~~~~~~^^^^^^^
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/vending_machine.py", line 5, in slot_position
    |     row = ord(code[0].upper()) - ord('A')
    |               ~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 20, in test_slot_posit
ion_no_crash
    |     assert False, f"slot_position crashed with {value!r}: {error}"
    | AssertionError: slot_position crashed with '': string index out of range
    | assert False
    | Failing test case: test_slot_position_no_crash(
    |     value='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py:19
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 18, in test_slot_posit
ion_no_crash
    |     slot_position(value)
    |     ~~~~~~~~~~~~~~^^^^^^^
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/vending_machine.py", line 5, in slot_position
    |     row = ord(code[0].upper()) - ord('A')
    |               ~~~~^^^
    | TypeError: 'NoneType' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 20, in test_slot_posit
```

Screenshot 6



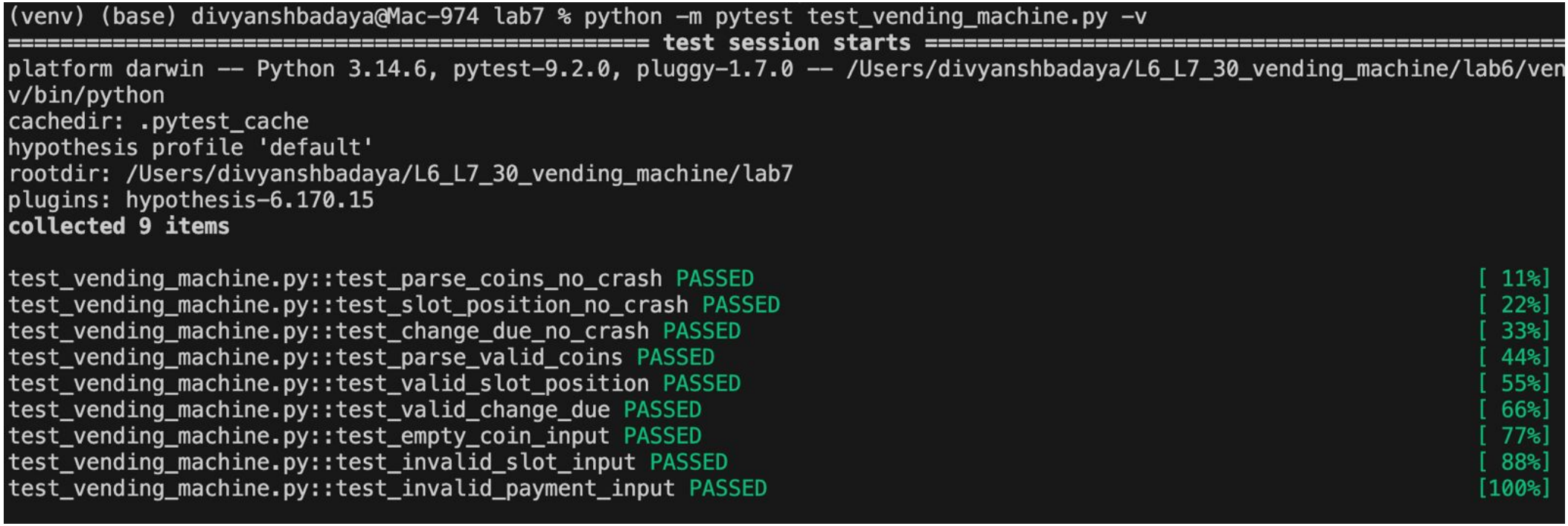
```
ion_no_crash
    |     assert False, f"slot_position crashed with {value!r}: {error}"
    | AssertionError: slot_position crashed with None: 'NoneType' object is not subscriptable
    | assert False
    | Failing test case: test_slot_position_no_crash(
    |     value=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py:19
    +-----------------------------------
_____________________________________ test_change_due_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 24, in test_change_due_n
o_crash
  |     def test_change_due_no_crash(value):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 26, in test_change_due
_no_crash
    |     change_due(value)
    |     ~~~~~~~~~~^^^^^^^
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/vending_machine.py", line 10, in change_due
    |     paid, price = payment.split('/')
    |     ^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 28, in test_change_due
```

Screenshot 7



```
_no_crash
    |     assert False, f"change_due crashed with {value!r}: {error}"
    | AssertionError: change_due crashed with '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_change_due_no_crash(
    |     value='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 26, in test_change_due
_no_crash
    |     change_due(value)
    |     ~~~~~~~~~~^^^^^^^
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/vending_machine.py", line 10, in change_due
    |     paid, price = payment.split('/')
    |                   ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/divyanshbadaya/L6_L7_30_vending_machine/lab7/test_vending_machine.py", line 28, in test_change_due
_no_crash
    |     assert False, f"change_due crashed with {value!r}: {error}"
    | AssertionError: change_due crashed with None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_change_due_no_crash(
    |     value=None,  # or any other generated value
    | )
    +------------------------------------
=========================================== short test summary info ============================================
FAILED test_vending_machine.py::test_parse_coins_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 su
b-exceptions)
FAILED test_vending_machine.py::test_slot_position_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3
sub-exceptions)
FAILED test_vending_machine.py::test_change_due_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub
-exceptions)
============================================== 3 failed in 0.77s ===============================================
```

Screenshot 8



```
(venv) (base) divyanshbadaya@Mac-974 lab7 % python -m pytest test_vending_machine.py -v
============================================== test session starts ==============================================
platform darwin -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0 -- /Users/divyanshbadaya/L6_L7_30_vending_machine/lab6/ven
v/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/divyanshbadaya/L6_L7_30_vending_machine/lab7
plugins: hypothesis-6.170.15
collected 9 items

test_vending_machine.py::test_parse_coins_no_crash PASSED                                                [ 11%]
test_vending_machine.py::test_slot_position_no_crash PASSED                                              [ 22%]
test_vending_machine.py::test_change_due_no_crash PASSED                                                 [ 33%]
test_vending_machine.py::test_parse_valid_coins PASSED                                                   [ 44%]
test_vending_machine.py::test_valid_slot_position PASSED                                                 [ 55%]
test_vending_machine.py::test_valid_change_due PASSED                                                    [ 66%]
test_vending_machine.py::test_empty_coin_input PASSED                                                    [ 77%]
test_vending_machine.py::test_invalid_slot_input PASSED                                                  [ 88%]
test_vending_machine.py::test_invalid_payment_input PASSED                                               [100%]
```

Screenshot 9

